Extended schedulability tests (1/3)

aggiungo poi il tempo di blocco che un task può subire dai task a priorità più basssa.

i task che hanno priorità più alta computeranno sempre

cioè che non sono indipendenti ma che condividono risorse tra loro

- Utilization based analysis
  - LL test for RM:
    A set of periodic tasks $\{\tau_1, \ldots, \tau_n\}$ with blocking factors and with $D_i = T_i$
    $\forall$ task τ_i is schedulable by RM if $\sum_{k|P_k>P_i} \frac{C_k}{T_k} + \frac{C_i+B_i}{T_i} \leq i(2^{1/i} - 1)$ $\forall$ task τ_i
  - HB test for RM:
    A set of periodic tasks $\{\tau_1, \ldots, \tau_n\}$ with blocking factors and with $D_i = T_i$
    $\forall$ task τ_i is schedulable by RM if $\prod_{k|P_k>P_i} \left(\frac{C_k}{T_k} + 1\right)\left(\frac{C_i+B_i}{T_i} + 1\right) \leq 2$ $\forall$ task τ_i
  - LL test for EDF:
    A set of periodic tasks $\{\tau_1, \ldots, \tau_n\}$ with blocking factors and with $D_i = T_i$
    $\forall$ task τ_i is schedulable by EDF if $\sum_{k|P_k>P_i} \frac{C_k}{T_k} + \frac{C_i+B_i}{T_i} \leq 1$ $\forall$ task τ_i

Extended schedulability tests (2/3)

- Response time analysis
  - A set of periodic tasks $\{\tau_1, \ldots, \tau_n\}$ with blocking factors and with $D_i \leq T_i$
    $\forall$ task τ_i is schedulable by DM if $R_i = C_i + B_i + \sum_{k|P_k > P_i} \left\lceil \frac{R_i}{T_k} \right\rceil C_k \leq D_i$ $\forall$ task τ_i
  - Iterative solution to compute R_i

$$\begin{cases} R_i^{(0)} = \sum_{k|P_k > P_i} C_k + C_i + B_i \\ R_i^{(j)} = C_i + B_i + \sum_{k|P_k > P_i} \left\lceil \frac{R_i^{(j-1)}}{T_k} C_k \right\rceil \end{cases}$$

Extended schedulability tests (3/3)

- Processor demand analysis
  - A set of periodic tasks $\{\tau_1, \ldots, \tau_n\}$ with blocking factors and with $D_i \leq T_i$ $\forall$ task τ_i is schedulable by EDF if $U < 1$ and dbf$(t)+B(t) \leq t \ \forall\, t \in D$
    - dbf$(t) = \sum_i \left\lfloor \frac{t + T_i - D_i}{T_i} C_i \right\rfloor$
    - $B(t) = \max_{i,j \mid i \neq j} \{\beta_{ij} \mid D_i > t \wedge D_j \leq t\}$ is termed **blocking function** (maximum time for which τ_i with $D_i \leq T_i$ may be blocked by τ_j with $D_i > t$)
    - β_{ij} := maximum time for which τ_i holds a resource that is also needed by τ_j
    - $D := \{d_i \mid d_i \leq \max\{D_{\max}, \min\{H, t^\star\}\}\}$, $D_{\max} := max_i\{D_i\}$
    - H :=lcm$(T_1, \ldots, T_n)$, $t^\star = \sum_i (T_i - D_i)U_i/(1 - U)$